

A geometric block-polar proof of the mixed-ball volume factorization

Candidate full solution to a closing question of arXiv:1505.05867

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Status. candidate full solution (likely valid; human review requested).

Source problem

Kempka and Vybírál [1] prove an exact gamma-function formula for the volume of the unit ball $B_{p,q}^{m,n}$ of $\ell_q^n(\ell_p^m)$. In their closing remarks (source PDF page 13), they observe that

$$\text{vol}(B_{p,q}^{m,n}) = \text{vol}(B_q^{mn}) \left(\frac{\text{vol}(B_p^m)}{\text{vol}(B_q^m)} \right)^n \quad (1)$$

and ask whether this has a geometric or combinatorial interpretation giving a non-analytic proof. We give such an interpretation and prove a more general star-body identity without evaluating any beta or gamma integral.

Transcription of the question

Does formula (1) admit a geometric or combinatorial interpretation which yields a non-analytic proof of the mixed-ball volume formula?

Source evidence

The two closing questions and formula (19), source PDF page 13.

$m = 1$ or $n = 1$: In these cases we directly see

$$B_{p,q}^{1,n}(\mathbb{R}) = B_q^n(\mathbb{R}) \quad \text{as well as} \quad B_{p,q}^{m,1}(\mathbb{R}) = B_p^m(\mathbb{R})$$

and Proposition 2 then coincides with 5 in Theorem 3

4 Closing remarks

The elements of the space $\mathbb{R}^{n \times m}$ are usually identified with $n \times m$ matrices. The most important matrix norms (like the nuclear norm or the spectral norm) are, however, not given by simple conditions on the entries of the matrix but rather on its singular values. We refer to [18] for results on volumes of unit balls with respect to these norms.

Theorem 3 gives the volume of the unit ball of $\ell_q^n(\ell_p^m)$. On the other hand, it leaves a number of questions open. Let us briefly discuss the most important ones.

- On many occurrences in mathematics it is more convenient to work with Lorentz sequence spaces instead of Lebesgue sequence spaces. If $0 < p \leq \infty$, the weak- ℓ_p^m (quasi-)norm is defined as

$$\|x\|_{p,\infty} = \max_{j=1,\dots,m} j^{1/p} x_j^*,$$

where $(x_j^*)_{j=1}^m$ is the non-increasing rearrangement of $x \in \mathbb{R}^m$. To our best knowledge, no closed formula for the volume of the unit ball $\{x \in \mathbb{R}^m : \|x\|_{p,\infty} \leq 1\}$ is available in the literature. The same holds true for the general Lorentz sequence spaces $\ell_{p,q}^m$ with $0 < q < \infty$ if $q \neq p$.

- Comparing 5 with 4, we observe that

$$\text{vol}(B_{p,q}^{m,n}(\mathbb{R})) = \text{vol}(B_q^{n,m}(\mathbb{R})) \cdot \frac{[\text{vol}(B_p^m(\mathbb{R}))]^n}{[\text{vol}(B_q^m(\mathbb{R}))]^n}. \quad (19)$$

This formula appeared implicitly already in [14]. We leave it open if this formula has some geometric or combinatoric interpretation, which would allow for an non-analytical proof of Theorem 3

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References

- [1] F. R. Bach, *Consistency of the group lasso and multiple kernel learning*, J. Mach. Learn. Res. **9** (2008), 1179–1225.

Geometric idea

For a star body $K \subset \mathbb{R}^m$, a gauge shell at radius r has volume $m \operatorname{vol}(K) r^{m-1} dr$. An outer ℓ_q constraint on n blocks sees only their gauge radii, so the admissible radial region is independent of the angular shape of K . Changing the inner body from L to K therefore multiplies each of the n shell measures by $\operatorname{vol}(K)/\operatorname{vol}(L)$.

Let K be a bounded Borel star body containing the origin in its interior, with gauge

$$\rho_K(x) = \inf\{t > 0 : x \in tK\}.$$

For $0 < q \leq \infty$, define its n -fold ℓ_q -sum by

$$\mathcal{B}_q^{(n)}(K) = \left\{ (x_1, \dots, x_n) \in (\mathbb{R}^m)^n : \|(\rho_K(x_1), \dots, \rho_K(x_n))\|_{\ell_q^n} \leq 1 \right\}.$$

Theorem 1. *For any two such star bodies $K, L \subset \mathbb{R}^m$, any $n \geq 1$, and $0 < q \leq \infty$,*

$$\frac{\operatorname{vol}_{mn}(\mathcal{B}_q^{(n)}(K))}{\operatorname{vol}_{mn}(\mathcal{B}_q^{(n)}(L))} = \left(\frac{\operatorname{vol}_m(K)}{\operatorname{vol}_m(L)} \right)^n.$$

Convexity is unnecessary.

Proof. Homothety of gauge sublevels gives, for $r \geq 0$,

$$\operatorname{vol}_m\{x : \rho_K(x) \leq r\} = \operatorname{vol}_m(rK) = r^m \operatorname{vol}_m(K).$$

Thus the pushforward of Lebesgue measure under ρ_K is

$$(\rho_K)_\#(dx) = m \operatorname{vol}_m(K) r^{m-1} dr.$$

Applying this identity independently in all n blocks shows that, for every nonnegative Borel function F ,

$$\begin{aligned} & \int_{(\mathbb{R}^m)^n} F(\rho_K(x_1), \dots, \rho_K(x_n)) dx_1 \cdots dx_n \\ &= (m \operatorname{vol}_m(K))^n \int_{[0, \infty)^n} F(r_1, \dots, r_n) \prod_{j=1}^n r_j^{m-1} dr_1 \cdots dr_n. \end{aligned}$$

Take F to be the indicator of

$$R_{q,n} = \{r \in [0, \infty)^n : \|r\|_{\ell_q^n} \leq 1\}.$$

Then

$$\operatorname{vol}_{mn}(\mathcal{B}_q^{(n)}(K)) = (m \operatorname{vol}_m(K))^n I_{m,n,q}, \quad I_{m,n,q} = \int_{R_{q,n}} \prod_{j=1}^n r_j^{m-1} dr.$$

The same constant occurs for L , so division proves the identity. When $q = \infty$, the identical argument uses $R_{\infty,n} = [0, 1]^n$. $\square$

A direct geometric transport

The cancellation above can be realized by an actual map. For a centrally symmetric star body K , write $S_K = \{\theta : \rho_K(\theta) = 1\}$ and let μ_K be normalized cone measure. The shell formula is the disintegration

$$dx = m \operatorname{vol}(K) r^{m-1} dr d\mu_K(\theta), \quad x = r\theta.$$

For $m \geq 2$, the nonatomic standard probability spaces (S_L, μ_L) and (S_K, μ_K) admit a measure-preserving bijection $\phi : S_L \rightarrow S_K$ modulo null sets. (For $m = 1$, match the two endpoints.) Its homogeneous extension

$$T(r\theta) = r\phi(\theta)$$

preserves gauge radius, $\rho_K(Tx) = \rho_L(x)$, and scales volume by $\operatorname{vol}(K)/\operatorname{vol}(L)$. Hence T^n maps $\mathcal{B}_q^{(n)}(L)$ onto $\mathcal{B}_q^{(n)}(K)$ modulo null sets while scaling volume by $(\operatorname{vol}(K)/\operatorname{vol}(L))^n$. This gives a literal geometric transport behind the factorization.

Resolution of the source question

Take $K = B_p^m$ and $L = B_q^m$. Then

$$\mathcal{B}_q^{(n)}(B_p^m) = B_{p,q}^{m,n}, \quad \mathcal{B}_q^{(n)}(B_q^m) = B_q^{mn}.$$

Theorem 1 is precisely (1). Geometrically, both bodies have the same weighted block-radius base $R_{q,n}$; replacing every inner q -ball by an inner p -ball changes only the n angular shell factors. This proves the factorization without calculating the common radial integral.

The argument includes all $0 < p, q \leq \infty$: neither the triangle inequality nor convexity is used, only positive homogeneity. At $q = \infty$ it reduces to the immediate product identity $\operatorname{vol}((B_p^m)^n) = \operatorname{vol}(B_p^m)^n$.

Verification and novelty bounds. The proof uses only $\{\rho_K \leq r\} = rK$, scaling of Lebesgue measure, and Tonelli’s theorem. Checks at $n = 1$, $m = 1$, $p = q$, and $q = \infty$ give the expected identities. No formula for $I_{m,n,q}$ is used, so there is no circular appeal to the source theorem.

A bounded search on 11 August 2026 covered the exact question and formula, the source’s citations, and combinations of “geometric interpretation”, “mixed-norm unit ball”, “volume ratio”, “block polar”, and “cone measure”. No later source giving this exact shell proof was found. The source’s separate Lorentz-volume question was answered later by Doležalová and Vybíral [2]; it is not claimed here. A negative search supports but cannot establish novelty.

Human-review recommendation. Verify the gauge-pushforward identity with the preferred harmless boundary convention, and confirm that the source’s requested “non-analytical” proof includes a gamma-free geometric shell disintegration. The volume identity itself then follows immediately.

References

- [1] H. Kempka and J. Vybíral, *Volumes of unit balls of mixed sequence spaces*, arXiv:1505.05867; [arXiv record](#).
- [2] A. Doležalová and J. Vybíral, *On the volume of unit balls of finite-dimensional Lorentz spaces*, arXiv:1906.04997; [arXiv record](#).